

YEAR 10

MATHS & SCIENCE

The GCSE learning year: building secure subject knowledge, application and practical competence

Government-source edition • Department for Education / GOV.UK only

Prepared for families supporting pupils through the first full GCSE teaching cycle.

1. How to use this guide

The DfE does not prescribe a national Year 10 topic order. It specifies Key Stage 4 programmes of study and GCSE subject content; awarding organisations turn that content into specifications, and schools decide the sequence across Years 10 and 11.

The Year 10 principle
Build the course before trying to revise the course. Year 10 should prioritise secure first learning, cumulative retrieval, practical competence and early repair of gaps.

2. GCSE Mathematics - the government content map

DOMAIN	GCSE CONTENT FRAMEWORK
NUMBER	Structure and calculation; fractions, decimals and percentages; measures and accuracy; powers, roots, standard form and, for higher attainment, surds and fractional indices.
ALGEBRA	Notation, vocabulary and manipulation; formulae; equations and inequalities; sequences; graphs; coordinate geometry; functions and, at higher tier, more advanced algebraic reasoning.
RATIO, PROPORTION & RATES	Ratios and fractions; scale; direct and inverse proportion; percentage change; growth and decay; compound measures including speed, density, pressure and rates of change.
GEOMETRY & MEASURES	Properties and constructions; angle reasoning; transformations; mensuration; similarity and

congruence; vectors; Pythagoras; trigonometry; circles; bearings and loci.

PROBABILITY	Single and combined events; relative frequency; sample spaces; tree diagrams and conditional probability at the appropriate level.
STATISTICS	Sampling; tables and charts; averages and spread; cumulative frequency and box plots where appropriate; scatter graphs; interpreting and comparing distributions.

3. Mathematics - what Year 10 should increasingly look like

PRIORITY	WHAT STRONG PROGRESS LOOKS LIKE
FLUENCY	Accurate number, algebra and calculator work; methods are quick enough not to overload harder problems.
REPRESENTATION	Moving confidently between algebra, graphs, tables, diagrams and words.
REASONING	Explaining steps, making deductions, checking plausibility and constructing arguments.
PROBLEM SOLVING	Choosing methods for multi-step and unfamiliar contexts rather than waiting for a topic cue.
CUMULATIVE RETRIEVAL	Older topics remain active while new topics are taught.
TIER AWARENESS	Foundation/Higher decisions are based on secure performance and school evidence, not aspiration alone.

4. GCSE Science - the government content map

DOMAIN	GCSE CONTENT FRAMEWORK
BIOLOGY	Cell biology; organisation; infection and response; bioenergetics; homeostasis and response; inheritance, variation and evolution; ecology.
CHEMISTRY	Atomic structure and periodic table; bonding, structure and properties; quantitative chemistry; chemical changes; energy changes; rates and equilibrium; organic chemistry; chemical analysis; atmosphere; using resources.

PHYSICS	Energy; electricity; particle model of matter; atomic structure; forces; waves; magnetism and electromagnetism; space physics where included in the relevant qualification.
PRACTICAL & ENQUIRY	Required practical experiences and apparatus techniques; planning, variables, measurement, data handling, graphing, evaluation, uncertainty and drawing evidence-based conclusions.
MATHEMATICAL SKILLS	Use of equations, ratios, percentages, standard form, graphs, gradients, areas, proportional reasoning, rearrangement and numerical analysis in scientific contexts.

Combined Science and separate Biology/Chemistry/Physics share a substantial common core, but separate sciences contain additional breadth/depth. The exact specification and sequencing depend on the school and awarding organisation.

5. Science - what Year 10 should increasingly look like

PRIORITY	WHAT STRONG PROGRESS LOOKS LIKE
KNOWLEDGE	Core facts, models and definitions are retrievable and connected across units.
APPLICATION	Students can use known science in unfamiliar contexts.
PRACTICALS	Students understand methods, variables, apparatus, data quality, risk and evaluation - not merely the “recipe”.
MATHEMATICS	Equations, rearrangement, graphs, standard form, percentages and proportional reasoning are routine in science.
EXPLANATION	Answers use linked scientific reasoning rather than isolated keywords.
CUMULATIVE REVIEW	Earlier biology, chemistry and physics remain in regular retrieval.

6. Assessment objectives families should understand

SUBJECT	WHAT ASSESSMENT REWARDS
MATHEMATICS	Use standard techniques; reason, interpret and

communicate mathematically; solve problems in mathematical and other contexts.

SCIENCE

Demonstrate knowledge and understanding; apply knowledge and understanding; analyse information and ideas, evaluate, make judgements and draw conclusions.

7. Parent support plan

- Know the exact exam board/specification used by the school once confirmed.
- Track taught topics and weak prerequisites, not just headline test scores.
- Protect weekly retrieval of older material so Year 10 knowledge does not decay.
- Treat practical science and mathematical science skills as examinable curriculum, not enrichment.
- Use school assessments diagnostically: identify whether marks were lost through knowledge, method, application, interpretation or exam execution.

8. What to obtain from your child's school

- Year 10 maths curriculum map and GCSE specification.
- Science pathway: Combined Science or separate sciences, plus specification.
- Foundation/Higher tier policy and review points.
- Required-practical schedule and how practical understanding is assessed.
- Internal assessment calendar, mock windows and expected independent-study routines.

Official government sources used

The curriculum content in this guide is derived from Department for Education / GOV.UK statutory programmes of study and GCSE subject-content publications. The year-specific sequencing lens is labelled as an Inspire support framework because government documents generally specify content by key stage or qualification, not by individual secondary-school year.

Department for Education. National curriculum in England: mathematics programme of study - key stage 4.

Ref: DFE-00496-2014

Published within the national curriculum mathematics programmes of study.

<https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study>

Department for Education. National curriculum in England: science programme of study - key stage 4.

Ref: DFE-00677-2014

Published 2 December 2014 within the national curriculum science programmes of study.

<https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study>

Department for Education. GCSE mathematics: subject content and assessment objectives.

Ref: DFE-00233-2013

Published 1 November 2013.

<https://www.gov.uk/government/publications/gcse-mathematics-subject-content-and-assessment-objectives>

Department for Education. GCSE subject content for combined science.

Ref: DFE-00351-2014

Published 9 April 2014; last updated 3 July 2015.

<https://www.gov.uk/government/publications/gcse-combined-science>

Department for Education. GCSE subject content for biology, chemistry and physics.

Ref: DFE-00352-2014

Published 9 April 2014; last updated 3 July 2015.

<https://www.gov.uk/government/publications/gcse-single-science>